National University of Sciences & Technology (NUST)
School of Electrical Engineering and Computer Science (SEECS)
Department of Computer Science

Department:	Faculty of Computing	**Knowledge Group:**	Core Computing
Program:	Software Engineering	**Class:**	BESE-15 (2k24) A
Course code:	CS-251	**Academic Session/Semester:**	Spring 2026
Course name:	Design and Analysis of Algorithms	**Pre/co requisite (course name and code, if applicable):**	CS-250 Data Structures and Algorithms
Credit hours:	3 + 0		

Course Synopsis	This course offers a comprehensive exploration of the design and analysis of algorithms, equipping students with the skills and knowledge needed to tackle real-world computational challenges. The course focuses on various algorithmic problem-solving techniques, including dynamic programming, divide-and-conquer, greedy algorithms and more, which are applied to a wide range of practical problems. Students learn to communicate their algorithmic solutions effectively, ensuring that they can present the algorithms with correctness proof.
Course Learning Outcomes (CLOs)	Upon completion of the course, students should demonstrate the ability to: • Analyze and classify algorithms of various types including divide-and-conquer algorithms, graph algorithms, dynamic programming, and linear programming, given problem descriptions and algorithmic specifications by comparing their design paradigms, computational complexity, and applicability to problem domains. • Construct algorithmic solutions to complex computational problems given problem statements with real-world constraints by synthesizing and integrating algorithmic paradigms, and producing a correct, efficient, and well-documented solution. • Assess and justify the run-time complexity of algorithms given algorithm descriptions and recurrence relations by applying asymptotic analysis techniques.
Course Schedule	Wednesday (CR # 16) (2:00pm – 4:00pm) Thursday (CR # 16) (12:00pm – 1:00pm)

Course Instructor	**Name**	**Office**	**Contact no.**	**E-mail**	**Office Hours**
	Dr. Zuhair Zafar	A-202	2166	zuhair.zafar@seecs.edu.pk	Wed. 4:00 – 5:00 pm

Mapping of the Course Learning Outcomes (CLO) to the Program Learning Outcomes (PLO), Bloom Taxonomy Domain & Level, Teaching & Learning (T&L) methods and Assessment methods:

No.	Course Learning Outcomes (verb, condition, standard)	PLO (SE)	PLO (CS)	BT Level	Teaching & Learning Methods	Assessment Methods
CLO 1	**Analyze** and **classify** algorithms of various types including divide-and-conquer algorithms, graph algorithms, dynamic programming, and linear programming, given problem descriptions and algorithmic specifications by comparing their design paradigms, computational complexity, and applicability to problem domains.	PLO 2	PLO 3	C-4 (Analysis)	Active learning, Cooperative Learning, Blended Learning	Assignment Project Quiz MSE ESE
CLO 2	**Construct** algorithmic solutions to complex computational problems given problem statements with real-world constraints by synthesizing and integrating algorithmic paradigms, and producing a correct, efficient, and well-documented solution.	PLO 3	PLO 4	C-6 (Create)	Active learning, Cooperative Learning, Blended Learning	Assignment Project MSE ESE

CLO 3	**Assess** and **justify** the run-time complexity of algorithms given algorithm descriptions and recurrence relations by applying asymptotic analysis techniques.	PLO 4	PLO 3	C-5 (Evaluate)	Active learning, Cooperative Learning, Blended Learning	Assignment Project Quiz MSE ESE

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities.
2.	Cooperative learning	Conducted through semester project. Students in a team of three will be given a project that requires design of algorithmic solutions. Compliance to the design specifications need to be given in the form of written reports.
3.	Blended learning	Conducted through Learning Management System (LMS) of NUST. All information as well as materials related to teaching and learning activities will be shared with class through this system. Some formative assessments will be also conducted using this system.

Weekly Schedule:

	Topic	
Week 1	• Role of Algorithms in Computing • Analysis of algorithms • Growth of Functions	
Week 2	**Recurrence Relations** • Solving recurrences using Substitution • Solving recurrences using Recursion Tree Method • Solving recurrences using Master Method	
Week 3	**Sorting Algorithms** • Sorting Algorithms: Selection Sort, Bubble Sort, Insertion Sort • Sorting Algorithms: Quick Sort • Analysis of Sorting Algorithms: Randomized Quick Sort	
Week 4	• Analysis of Sorting Algorithms: Merge Sort • Analysis of Sorting Algorithms: Heap Sort • Priority Queues (Heaps)	Assign # 1
Week 5	• Analysis of Sorting Algorithms: Counting Sort • Analysis of Sorting Algorithms: Radix Sort • Analysis of Sorting Algorithms: Bucket Sort	
Week 6	**Graph Algorithms** • Introduction to Graphs • Breadth-First Search and Depth-First Search • Introduction to Trees, Minimum Spanning Trees, Kruskal's Algorithm	
Week 7	• Prim's algorithm, Correctness, Analysis **Shortest Path Algorithms** • Analysis of Dijkstra shortest path algorithm	Assign # 2
Week 8	**Mid Semester Exam (MSE)**	

Week 9	• Analysis of Bellman-Ford shortest path algorithm • Analysis of Floyd-Warshall all pair shortest path algorithm • Analysis of Johnson all pair shortest path algorithm	
Week 10	**Greedy algorithms** • Huffman coding • Activity selection problem • Bin Packing Problem • Money Counting Problem	
Week 11	**Linear Programming** • System of linear inequalities • Simplex Algorithm	Semester Project
Week 12	**String Matching Algorithms** • Naïve String-Matching Algorithm • Rabin Karp String Matching Algorithm • Knuth-Morris-Pratt String Matching Algorithm	Assign # 3
Week 13	**Memoization** **Dynamic programming** • Knapsack algorithm	
Week 14	• Weighted Interval Scheduling Problem • Assembly Line Scheduling	
Week 15	**NP Completeness**	
Week 16	**Approximate Algorithms**	
Week 17	**Project Presentations**	
Week 18	**End Semester Exam (ESE)**	

Assessment Methods:

Course Assessment Weightages (In accordance with NUST statutes)	
Quizzes (10-15%)	**15%**
Assignments (5-10%)	**5%**
Semester Project (0-20%)	**10%**
Mid Semester Exam (MSE) (25-35%)	**25%** **(Duration: 2 Hours)**
End Semester Exam (ESE) (40-50%)	**45%** **(Duration: 3 Hours)**
Total	**100%**

Learning Resources:

Books	
Textbooks	1. Thomas H. Cormen, Introduction to Algorithms (4th edition) 2022, MIT Press. 2. Jon Kleinberg, Eva Tardos. Algorithm Design, 1st Edition. 2006. Pearson Education, Inc.
Reference Materials	• Clifford A. Shaffer, Data Structures and Algorithm Analysis, Edition 3.2, 2012. • Steven S Skiena, The Algorithm Design Manual, 2nd Edition (2008). • M. A. Iqbal, Graph Theory & Algorithms, Google Books, 2012. • Online Resources (will be shared on the LMS)

Course Assessments:

Exam:	1 Mid Semester Exam (MSE) and 1 End Semester Exam (ESE)
Homework:	3 Assignments
Quizzes:	6 Quizzes
Semester Project	1 Semester Project

Grading Policy:

Quiz Policy:	The quizzes will be unannounced and will be conducted at the start of the lecture and normally last for ten minutes. The question framed is to test the concepts involved in the last few lectures. Number of quizzes that will be used for evaluation is at the instructor's discretion.
Assignment Policy:	In order to develop a comprehensive understanding of the subject, assignments will be given. Late assignments will not be accepted / graded. All assignments will count towards the total (No 'best-of' policy). The students are advised to do the assignment themselves. Copying assignments is highly discouraged and violations will be dealt with severely by referring any occurrences to the disciplinary committee. The questions in the assignment are meant to be challenging to give students confidence and extensive knowledge about the subject matter and enable them to prepare for the exams.
Project Policy: **Working in Groups and Grading**	Students are required to develop a project during the course which should be completed towards the end of the semester. Students will be graded based on project deliverables and presentation at the end. Students are advised to work in a group/team for projects. Each member of the group is expected to contribute to project development. The students should explicitly indicate the individual member contribution during project presentation. The contribution by each student will be judged by QA session during presentation and viva afterwards.
Class Participation:	The students are encouraged to participate in class by 1) Answering the questions by the instructor, 2) Answering the question from fellow student(s) if given opportunity by the instructor, 3) Asking question from the instructor, when invited, 4) Participate in class discussion, 5) Participate in the in-class group activity, 6) Project presentation. The class participation will be observed by the instructor.
Plagiarism:	SEECS maintains a zero-tolerance policy towards plagiarism. While collaboration in this course is highly encouraged, you must ensure that you do not claim other people's work/ ideas as your own. Plagiarism occurs when the words, ideas, assertions, theories, figures, images, programming codes of others are presented as your own work. You must cite and acknowledge all sources of information in your assignments. Failing to comply with the SEECS plagiarism policy will lead to strict penalties including zero marks in assignments and referral to the academic coordination office for disciplinary action. More information on plagiarism policy is available at the following links. SEECS Plagiarism Policy: http://seecs.nust.edu.pk/Internal/downloads/ HEC Plagiarism Policy: https://www.hec.gov.pk/english/services/faculty/Plagiarism/Pages/default.aspx

Approved by:



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Knowledge Group Head